

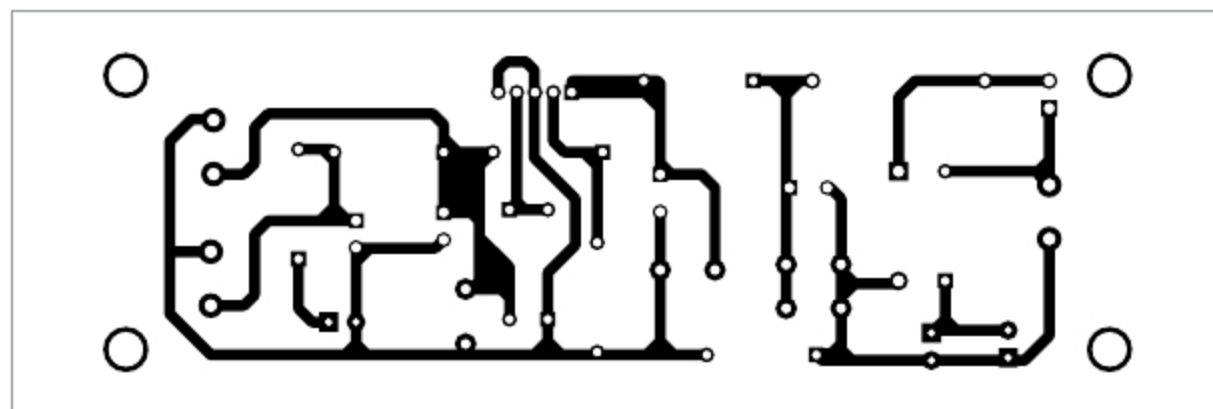


Fig. 2: Actual-size PCB layout of the 12V DC to 9V DC converter

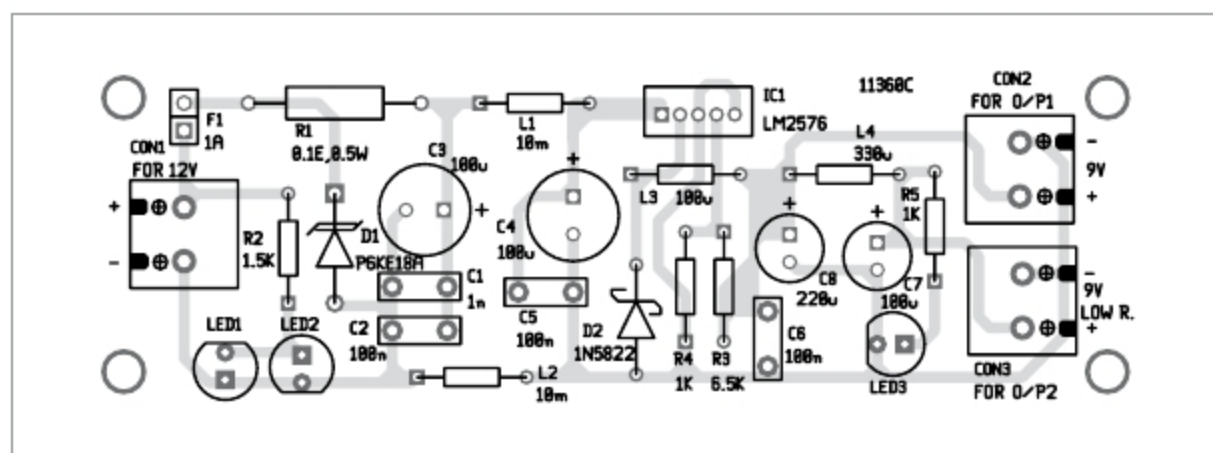


Fig. 3: Components layout for the PCB

LED3 on the front panel, and CON2 through CON3 on the rear side of the panel for outputs.

Heat-sink for LM2576 depends on input voltage and load current. Power dissipation on LM2576 can be significant and, hence, requires a small heat-sink with thermal resistance lower than 30°C/W.

Output voltage (9V, 3A) is available

PARTS LIST

Semiconductors:

- IC1 - LM2576 adjustable voltage regulator
- D1 - P6KE18A TVS diode
- D2 - 1N5822 Schottky diode
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon), unless stated otherwise:

- R1 - 1-ohm, 0.5W
- R2 - 1.5-kilo-ohm
- R3 - 6.5-kilo-ohm
- R4, R5 - 1-kilo-ohm

Capacitors:

- C1 - 10nF ceramic disk
- C2, C5, C6 - 100nF ceramic disk
- C3, C4, C7 - 100µF, 25V electrolytic
- C8 - 220µF, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- F1 - 3A fuse
- L1, L2 - 10mH inductor
- L3 - 100µH inductor
- L4 - 330µH inductor
- Heat-sink for LM2576
- 12V, 7Ah battery/car battery
- PCB

(All these are available from kitsnspares.com)

across connector CON2. Low-ripples 9V output voltage can be obtained at CON3. Input voltage from 11V to 15V can be connected at CON1. Maximum input voltage is limited by maximum voltage of input capacitors and maximum power dissipation of the IC.

The circuit can be connected to the electrical installation of your car, but make sure that input is within the working range (11V to 15V). Care should be taken because if the engine is running and you disconnect the 12V battery of the car, or if the voltage regulator in the car is not working properly, voltage generated from the generator could be much more than 40V and pulses may go up to around 100V.

Also, transitions can generate negative input voltages. In real applications, input filtering capacitors should have working voltages of 150V. It is suggested to have a set of electrolytic and ceramic capacitors covering large frequency ranges. **EFY**



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-Morris Hite

